Lab Title: Introduction to Docker Swarm

Lab Description: In this lab, you will learn the basics of Docker Swarm, a native clustering and orchestration solution for Docker. Docker Swarm allows you to create and manage a swarm of Docker nodes, turning them into a distributed system that can run and scale containers across multiple machines. By the end of this lab, you will have hands-on experience with setting up a Docker Swarm, deploying services, and managing the swarm cluster.

Lab Environment:

- Docker installed on the host machine
- Access to a terminal or command prompt

Lab Tasks:

1. Initialize a Docker Swarm:
    - Open a terminal or command prompt.
    - Run the following command to initialize a Docker Swarm:

docker swarm init

- Take note of the generated command to add worker nodes to the swarm.

2. Join Worker Nodes to the Swarm:
    - On another machine or virtual machine, open a terminal or command prompt.
    - Run the command obtained from the previous step to join the worker node to the swarm.
    - Repeat this step for additional worker nodes if available.
3. Deploy a Service:
    - Create a Docker service using a sample application.
    - Pull the "nginx" image from Docker Hub:

docker pull nginx

- Deploy the "nginx" service on the swarm:

docker service create --name my-nginx --replicas 3 -p 80:80 nginx

- Verify that the service is running and the replicas are spread across the swarm nodes:

docker service ls docker service ps my-nginx

4. Scale the Service:
    - Increase the number of replicas for the "my-nginx" service:

docker service scale my-nginx=5

- Verify that the additional replicas are created and distributed:

docker service ps my-nginx

5. Update the Service:

    - Update the service by changing the published port to 8080:

docker service update --publish-add 8080:80 my-nginx

- Verify that the service has been updated and the new port is being used:

docker service inspect --format='{{.Endpoint.Ports}}' my-nginx

6. Remove the Service and Leave Swarm:

    - Remove the "my-nginx" service:

docker service rm my-nginx

- On each worker node, leave the swarm:

docker swarm leave

Lab Conclusion: In this lab, you successfully set up a Docker Swarm, deployed a service, scaled it, and updated it. Docker Swarm provides a powerful way to manage containerized applications in a distributed environment. With this knowledge, you can explore more advanced features of Docker Swarm and build robust, scalable systems using Docker containers.